

Disentangling location and concentration effects in von Mises-Fisher regression

Kipoong Kim¹ Jörn Schulz² Sungkyu Jung³

¹Department of Statistics, Changwon National University

²Department of Mathematics and Physics, University of Stavanger

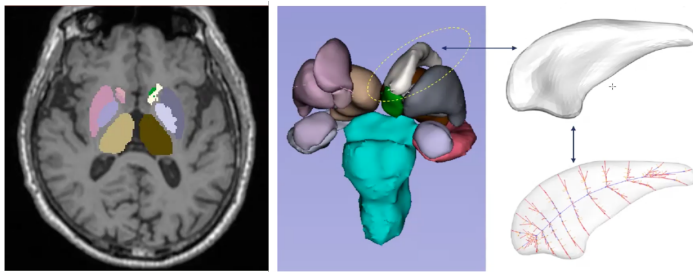
³Department of Statistics, Seoul National University

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Motivating data

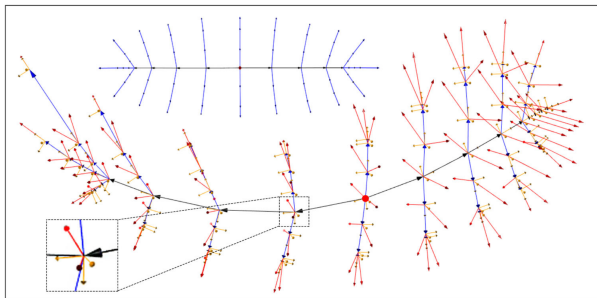
Shape data from brain MRI

- The ParkWest study has collected brain MRI data on patients with Parkinson's disease and on healthy controls.
- We aim to test whether the **shape of the hippocampus**, derived from the MRI, differs between the **Parkinson's disease** group and the **control** group, after adjusting for age, sex, and other covariates.



Locally parametrized discrete skeletal representation

- To represent the shape mathematically, we use the LP-ds-rep of Taheri and Schulz (2023), under which each hippocampus becomes a set of frames attached to an interior skeleton.
- The representation gives the features that capture the shape: the **centroid**, the **connection vectors** forming the spine, the **spoke vectors** extending toward the boundary, and the **frames**.



The sample space

- Each connection and each spoke is a direction together with a length, so each shape is a point in

$$\underbrace{\mathbb{R}^3}_{\text{centroid}} \times \underbrace{(\mathbb{S}^2)^{n_c} \times (\mathbb{R}_+)^{n_c}}_{\text{connections}} \times \underbrace{(\mathbb{S}^2)^{n_s} \times (\mathbb{R}_+)^{n_s}}_{\text{spokes}} \times \underbrace{(\text{SO}(3))^{n_c}}_{\text{frames}},$$

where n_c and n_s are the numbers of connections and spokes.

- ▶ $\mathbb{S}^{q-1} := \{\mathbf{y} \in \mathbb{R}^q : \mathbf{y}^T \mathbf{y} = 1\}$
- ▶ $\text{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} : \mathbf{R}^T \mathbf{R} = \mathbf{I}, |\mathbf{R}| = 1\}$ is the special orthogonal group, which can be represented by a unit quaternion in $\mathbb{S}^3$.
- Almost every component lives on a sphere.
- In this work, we focus on the **connection direction** $\mathbb{S}^2$ at each region, and ask how it changes with covariates such as disease status, age, and sex.

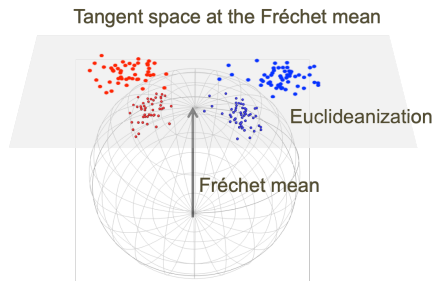
One approach: Euclideanization

- Compute the Fréchet mean $\bar{\mathbf{y}}$ of $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{S}^{q-1}$ and map the responses onto the tangent space at $\bar{\mathbf{y}}$ by the log map,

$$\mathbf{v}_i = \text{Log}_{\bar{\mathbf{y}}}(\mathbf{y}_i) \in T_{\bar{\mathbf{y}}}\mathbb{S}^{q-1} \cong \mathbb{R}^{q-1},$$

then apply an ordinary Euclidean regression to $\mathbf{v}_1, \dots, \mathbf{v}_n$.

- **Limitation.** The log map preserves geodesic distances only from $\bar{\mathbf{y}}$, not the **pairwise** distances among the points, so the distortion grows as the data spread away from the mean.

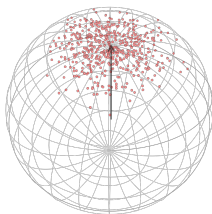


The von Mises-Fisher distribution

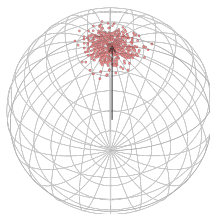
- For $\mathbf{y} \in \mathbb{S}^{q-1}$, the vMF density is

$$f_{\text{vMF}}(\mathbf{y}; \boldsymbol{\zeta}, \kappa) = C_q(\kappa) \exp(\kappa \boldsymbol{\zeta}^\top \mathbf{y}), \quad C_q(\kappa) = \frac{\kappa^{q/2-1}}{(2\pi)^{q/2} I_{q/2-1}(\kappa)},$$

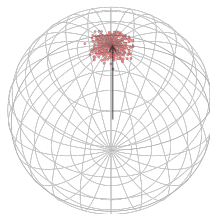
with mean direction $\boldsymbol{\zeta} \in \mathbb{S}^{q-1}$, concentration $\kappa \geq 0$, and $I_\nu(\cdot)$ the modified Bessel function of the first kind of order ν .



(a) $\kappa = 5$



(b) $\kappa = 10$



(c) $\kappa = 15$

Figure 1: $q = 3$ and $\boldsymbol{\zeta} = (0, 0, 1)^\top$

A covariate can act in two ways

Three hypotheses on the sphere

- A location effect, a concentration effect, and their combination.

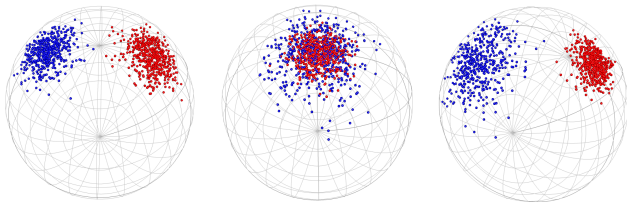


Figure 2: Binary covariate: the two groups appear as control (red) and case (blue) clusters

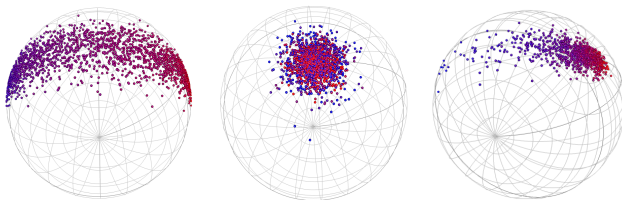


Figure 3: Continuous covariate: the response sweeps along the sphere as x_i varies, colored from red (low) to blue (high)

Reparameterization by the natural parameter

- We instead use the natural parameter $\boldsymbol{\theta} = \kappa \boldsymbol{\zeta} \in \mathbb{R}^q$, which carries the direction and the concentration in a single unconstrained vector.

$$\boldsymbol{\zeta} = \boldsymbol{\theta} / \|\boldsymbol{\theta}\|, \quad \kappa = \|\boldsymbol{\theta}\|$$

- The reparametrized vMF density takes **the canonical exponential family form**

$$f_{\text{vMF}}(\mathbf{y}; \boldsymbol{\theta}) = \exp\{\boldsymbol{\theta}^\top \mathbf{y} - b(\boldsymbol{\theta})\}, \quad b(\boldsymbol{\theta}) = -\log C_q(\|\boldsymbol{\theta}\|).$$

- For $\boldsymbol{\theta} \neq \mathbf{0}$, the first derivative of b is the mean vector

$$\mathbb{E}(\mathbf{y}) = \nabla b(\boldsymbol{\theta}) = A_q(\|\boldsymbol{\theta}\|) \frac{\boldsymbol{\theta}}{\|\boldsymbol{\theta}\|}, \quad A_q(\kappa) = \frac{I_{q/2}(\kappa)}{I_{q/2-1}(\kappa)},$$

where A_q , the mean resultant length, is strictly increasing from $[0, \infty)$ onto $[0, 1)$.

Linking the spherical response to the covariates

- The GLM for the reparameterized vMF distribution is

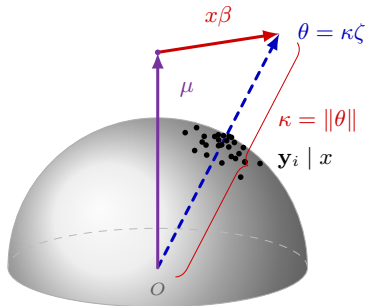
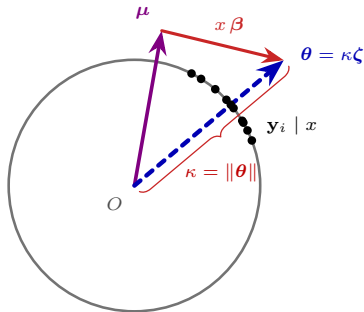
$$g(\mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i]) = \boldsymbol{\theta}_i = \boldsymbol{\mu} + \sum_{j=1}^p x_{ij} \boldsymbol{\beta}_j = \boldsymbol{\mu} + \mathbf{B}^\top \mathbf{x}_i, \quad \mathbf{y}_i \mid \mathbf{x}_i \sim \text{vMF}(\boldsymbol{\theta}_i),$$

where $\boldsymbol{\mu} \in \mathbb{R}^q$ is the baseline parameter, $\mathbf{B} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_p]^\top \in \mathbb{R}^{p \times q}$ the coefficient matrix, and the canonical link $g(\boldsymbol{\xi}) = A_q^{-1}(\|\boldsymbol{\xi}\|) \boldsymbol{\xi} / \|\boldsymbol{\xi}\|$, is the inverse of ∇b , defined on the open unit ball.

- Here, we assume that the covariates are column-centered so that $\boldsymbol{\mu}$ is the average of natural parameters.

- From the single vector θ_i we recover the **location** and the **concentration**,

$$\zeta_i = \frac{\theta_i}{\|\theta_i\|}, \quad \kappa_i = \|\theta_i\|, \quad \text{so that } \mathbf{y}_i \mid \mathbf{x}_i \sim \text{vMF}(\zeta_i, \kappa_i).$$



Projection-based Inference: Decomposing $\hat{\beta}_j$

- Two orthogonal projections

$$\hat{\beta}_j = \underbrace{\mathbf{P}_{\hat{\mu}} \hat{\beta}_j}_{\text{concentration effect}} + \underbrace{\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j}_{\text{location effect}}$$

where $\mathbf{P}_{\mu} = \mu\mu^\top / \|\mu\|^2$ and $\mathbf{P}_{\mu^\perp} = \mathbf{I}_q - \mathbf{P}_{\mu}$

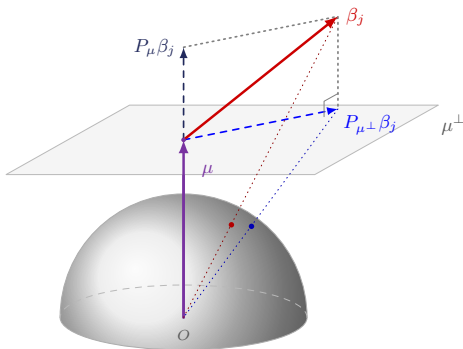


Figure 4: Decomposition in $\mathbb{R}^q$

Model estimation

Vectorized parameterization

- Let us express the parameters as

$$\boldsymbol{\gamma} = (\boldsymbol{\mu}^\top, \boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_p^\top)^\top \in \mathbb{R}^{(p+1)q}.$$

With $\tilde{\mathbf{z}}_i = (1, \mathbf{x}_i^\top)^\top \otimes \mathbf{I}_q \in \mathbb{R}^{(p+1)q \times q}$ and $\tilde{\mathbf{Z}} = (\tilde{\mathbf{z}}_1, \dots, \tilde{\mathbf{z}}_n)^\top \in \mathbb{R}^{nq \times (p+1)q}$, we have

$$\boldsymbol{\theta}_i = \boldsymbol{\mu} + \sum_{j=1}^p x_{ij} \boldsymbol{\beta}_j = \tilde{\mathbf{z}}_i^\top \boldsymbol{\gamma}.$$

- Log-likelihood:

$$\ell_n(\boldsymbol{\gamma}) = \sum_{i=1}^n \{ \boldsymbol{\gamma}^\top \tilde{\mathbf{z}}_i \mathbf{y}_i - b(\tilde{\mathbf{z}}_i^\top \boldsymbol{\gamma}) \}.$$

- Score and Fisher information:

$$\mathbf{s}_n(\boldsymbol{\gamma}) = \sum_{i=1}^n \tilde{\mathbf{z}}_i \{ \mathbf{y}_i - \nabla b(\tilde{\mathbf{z}}_i^\top \boldsymbol{\gamma}) \}, \quad \mathbf{F}_n(\boldsymbol{\gamma}) = \sum_{i=1}^n \tilde{\mathbf{z}}_i \nabla^2 b(\tilde{\mathbf{z}}_i^\top \boldsymbol{\gamma}) \tilde{\mathbf{z}}_i^\top.$$

Maximum likelihood estimation

- Since the family is canonical, the log-likelihood function is concave:

$$\hat{\gamma} = \arg \max_{\gamma \in \mathbb{R}^{(p+1)q}} \sum_{i=1}^n \{ \gamma^\top \tilde{\mathbf{z}}_i \mathbf{y}_i - b(\tilde{\mathbf{z}}_i^\top \gamma) \}.$$

Proposition 3.1 (Uniqueness of the MLE)

Let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times p}$. If $\mathbf{X}$ is of full rank under column-centering $\mathbf{1}_n \perp \text{col}(\mathbf{X})$, then $\mathbf{F}_n(\gamma) \succ \mathbf{0}$ for all $\gamma \in \mathbb{R}^{(p+1)q}$, and the MLE of γ is unique.

- $\hat{\gamma}$ can be computed via Iteratively Re-weighted Least Squares (IRLS):

$$\gamma^{(t+1)} = \left(\tilde{\mathbf{Z}}^\top \mathbf{W}^{(t)} \tilde{\mathbf{Z}} \right)^{-1} \tilde{\mathbf{Z}}^\top \mathbf{W}^{(t)} \mathbf{H}^{(t)},$$

where $\mathbf{W}^{(t)} = \text{blkdiag}\{\nabla^2 b(\tilde{\mathbf{z}}_i^\top \gamma^{(t)})\} \in \mathbb{R}^{nq \times nq}$ is block-diagonal,

$\mathbf{H}^{(t)} = \tilde{\mathbf{Z}} \gamma^{(t)} + (\mathbf{W}^{(t)})^{-1} \{\text{vec}(\mathbf{Y}^\top) - \boldsymbol{\eta}^{(t)}\} \in \mathbb{R}^{nq}$, and

$\boldsymbol{\eta}^{(t)} = (\nabla b(\tilde{\mathbf{z}}_1^\top \gamma^{(t)}), \dots, \nabla b(\tilde{\mathbf{z}}_n^\top \gamma^{(t)})) \in \mathbb{R}^{nq}$.

Regularity Assumptions (Fahrmeir & Kaufmann, 1985)

Assumption	Content and Role
A1 (Compact covariates)	Covariates lie in a compact set: $\{\mathbf{x}_i\} \subset K_x$ <i>Bounds the third moment of the score, together with $\ \mathbf{y}_i\ = 1$</i>
A2 (Interior truth)	γ_0 lies in the interior of an open convex $K_\gamma \subset \mathbb{R}^{(p+1)q}$ <i>Allows the Taylor expansion of the score at γ_0</i>
A3 (Growing information)	$\lambda_{\min}(\mathbf{F}_n(\gamma_0)) \geq n/c$ for some $c > 0$ <i>Makes $N_n(\epsilon)$ shrink at rate $O(n^{-1/2})$</i>
A4 (Stable limit)	$n^{-1}\mathbf{F}_n(\gamma_0) \rightarrow \mathbf{F}(\gamma_0) \succ \mathbf{0}$ <i>Gives the $\sqrt{n}$ scaling and a consistent variance estimator</i>

- A1–A3 suffice for consistency and asymptotic normality; A4 is needed only to express the limit in the $\sqrt{n}$ form.

Two key lemmas

- Since $\mathbf{F}_n(\gamma_0) \succ \mathbf{0}$, it admits the Cholesky factorization

$$\mathbf{F}_n(\gamma_0) = \mathbf{F}_n(\gamma_0)^{1/2} \mathbf{F}_n(\gamma_0)^{\top/2}$$

with $\mathbf{F}_n(\gamma_0)^{1/2}$ lower triangular, and we write $\mathbf{F}_n(\gamma_0)^{-1/2} = (\mathbf{F}_n(\gamma_0)^{1/2})^{-1}$ and $\mathbf{F}_n(\gamma_0)^{-\top/2} = (\mathbf{F}_n(\gamma_0)^{\top/2})^{-1}$.

- These factors normalize the information and the score over the neighborhood $N_n(\epsilon) = \{\gamma : \|\mathbf{F}_n(\gamma_0)^{\top/2}(\gamma - \gamma_0)\| \leq \epsilon\}$ of γ_0 , whose radius is of order $O(n^{-1/2})$ under A3.

Lemma 4.1 (Uniform convergence of the information ratio)

Under A1–A3, for every $\epsilon > 0$,

$$\max_{\gamma \in N_n(\epsilon)} \|\mathbf{V}_n(\gamma) - \mathbf{I}_{(p+1)q}\| \xrightarrow{p} 0, \quad \mathbf{V}_n(\gamma) := \mathbf{F}_n(\gamma_0)^{-1/2} \mathbf{F}_n(\gamma) \mathbf{F}_n(\gamma_0)^{-\top/2}.$$

Lemma 4.2 (Asymptotic normality of the score)

Under A1–A3,

$$\mathbf{F}_n(\gamma_0)^{-1/2} \mathbf{s}_n(\gamma_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{I}_{(p+1)q}).$$

Consistency and asymptotic normality

Theorem 4.3 (Under A1–3)

- (i) The MLE is weakly consistent, $\hat{\gamma} \xrightarrow{p} \gamma_0$.
- (ii) The MLE is asymptotically normal under A4,

$$\sqrt{n}(\hat{\gamma} - \gamma_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma), \quad \Sigma := \mathbf{F}(\gamma_0)^{-1} = \left\{ \lim_{n \rightarrow \infty} n^{-1} \mathbf{F}_n(\gamma_0) \right\}^{-1}.$$

- With selection matrices $\mathbf{D}_\mu, \mathbf{D}_j \in \mathbb{R}^{q \times (p+1)q}$ such that $\boldsymbol{\mu} = \mathbf{D}_\mu \boldsymbol{\gamma}$ and $\boldsymbol{\beta}_j = \mathbf{D}_j \boldsymbol{\gamma}$, this theorem yields the joint distribution of $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\beta}}_j$:

$$\sqrt{n} \begin{pmatrix} \hat{\boldsymbol{\mu}} - \boldsymbol{\mu}_0 \\ \hat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}_{j,0} \end{pmatrix} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_{\text{joint}}), \quad \Sigma_{\text{joint}} = \begin{pmatrix} \Sigma_{\mu\mu} & \Sigma_{\mu\beta_j} \\ \Sigma_{\beta_j\mu} & \Sigma_{\beta_j\beta_j} \end{pmatrix},$$

where $\Sigma_{\mu\mu} = \mathbf{D}_\mu \Sigma \mathbf{D}_\mu^\top$, $\Sigma_{\beta_j} = \mathbf{D}_j \Sigma \mathbf{D}_j^\top$, and $\Sigma_{\mu\beta_j} = \mathbf{D}_\mu \Sigma \mathbf{D}_j^\top = \Sigma_{\beta_j\mu}^\top$.

- In practice Σ is replaced by the plug-in estimator $\hat{\Sigma} = n \mathbf{F}_n(\hat{\gamma})^{-1}$, which is consistent by (i).

Asymptotic Distribution of Projected Estimators

Theorem 5.1

Suppose A1–A4 hold and $\mu_0 \neq \mathbf{0}$. Let $s_0 = \mu_0^\top \beta_{j,0} / \|\mu_0\|^2$ and $\mathbf{G}_0 = \mu_0 \beta_{j,0}^\top \mathbf{P}_{\mu_0^\perp} / \|\mu_0\|^2 + s_0 \mathbf{P}_{\mu_0^\perp}$. Applying the **delta method** to $g_\perp(\mu, \beta_j) = \mathbf{P}_{\mu^\perp} \beta_j$ and $g_\parallel(\mu, \beta_j) = \mathbf{P}_\mu \beta_j$ gives

(i) **Location effect:**

$$\sqrt{n}(\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j - \mathbf{P}_{\mu_0^\perp} \beta_{j,0}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega_{loc}),$$

with $\Omega_{loc} = \mathbf{J}_\perp \Sigma_{joint} \mathbf{J}_\perp^\top$ and $\mathbf{J}_\perp = [-\mathbf{G}_0, \mathbf{P}_{\mu_0^\perp}]$.

(ii) **Concentration effect:**

$$\sqrt{n}(\mathbf{P}_{\hat{\mu}} \hat{\beta}_j - \mathbf{P}_{\mu_0} \beta_{j,0}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega_{conc}),$$

with $\Omega_{conc} = \mathbf{J}_\parallel \Sigma_{joint} \mathbf{J}_\parallel^\top$ and $\mathbf{J}_\parallel = [\mathbf{G}_0, \mathbf{P}_{\mu_0}]$

- Ω_{loc} and Ω_{conc} are consistently estimated by their plug-in versions $\hat{\Omega}_{loc}$ and $\hat{\Omega}_{conc}$, obtained by replacing $(\mu_0, \beta_{j,0}, \Sigma_{joint})$ with $(\hat{\mu}, \hat{\beta}_j, \hat{\Sigma}_{joint})$.

Test statistics

- Corresponding to the three cases, we consider

$$H_0 : \beta_{j,0} = \mathbf{0}, \quad H_0^{\text{loc}} : \mathbf{P}_{\mu_0^\perp} \beta_{j,0} = \mathbf{0}, \quad H_0^{\text{conc}} : \mathbf{P}_{\mu_0} \beta_{j,0} = \mathbf{0}.$$

- Under H_0^{loc} and H_0^{conc} the asymptotic covariance is **singular**, so we consider $\hat{\mathbf{U}} \in \mathbb{R}^{q \times (q-1)}$ having orthonormal columns spanning $\hat{\mu}^\perp$ and

$$\hat{\Omega}_{\text{loc}}^+ = \hat{\mathbf{U}}(\hat{\mathbf{U}}^\top \hat{\Omega}_{\text{loc}} \hat{\mathbf{U}})^{-1} \hat{\mathbf{U}}^\top, \quad \hat{\Omega}_{\text{conc}}^+ = \frac{\hat{\mu} \hat{\mu}^\top}{\hat{\mu}^\top \hat{\Omega}_{\text{conc}} \hat{\mu}},$$

both invariant to the choice of $\hat{\mathbf{U}}$.

- The **Wald-type test statistics** are

$$W_{\text{global}} = n \hat{\beta}_j^\top \hat{\Sigma}_{\beta_j}^{-1} \hat{\beta}_j,$$

$$W_{\text{loc}} = n(\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j)^\top \hat{\Omega}_{\text{loc}}^+ (\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j), \quad W_{\text{conc}} = n(\mathbf{P}_{\hat{\mu}} \hat{\beta}_j)^\top \hat{\Omega}_{\text{conc}}^+ (\mathbf{P}_{\hat{\mu}} \hat{\beta}_j).$$

- Under A1–A4, each test statistic is asymptotically chi-squared under its null:

$$W_{\text{global}} \xrightarrow{d} \chi_q^2, \quad W_{\text{loc}} \xrightarrow{d} \chi_{q-1}^2, \quad W_{\text{conc}} \xrightarrow{d} \chi_1^2.$$

Type I error control and local power

Theorem 5.2 (Under A1–A4)

(i) Both tests attain the nominal level asymptotically,

$$\lim_{n \rightarrow \infty} P_{H_0^{loc}}(W_{loc} > \chi_{q-1, \alpha}^2) = \alpha, \quad \lim_{n \rightarrow \infty} P_{H_0^{conc}}(W_{conc} > \chi_{1, \alpha}^2) = \alpha.$$

(ii) Under the Pitman local alternatives, with $\delta_{\perp} \perp \mu_0$, $\delta_{\perp} \neq \mathbf{0}$, and $\delta_{\parallel} \neq 0$,

$$H_{1,n}^{loc} : \mathbf{P}_{\mu_0^{\perp}} \beta_{j,0} = \frac{\delta_{\perp}}{\sqrt{n}} \implies W_{loc} \xrightarrow{d} \chi_{q-1}^2(\lambda_{loc}),$$

$$H_{1,n}^{conc} : \mathbf{P}_{\mu_0} \beta_{j,0} = \frac{\delta_{\parallel} \mu_0}{\sqrt{n} \|\mu_0\|} \implies W_{conc} \xrightarrow{d} \chi_1^2(\lambda_{conc}),$$

where $\lambda_{loc} = \delta_{\perp}^{\top} \Omega_{loc}^{+} \delta_{\perp}$ and $\lambda_{conc} = \delta_{\parallel}^2 \mu_0^{\top} \Omega_{conc}^{+} \mu_0 / \|\mu_0\|^2$ are the noncentrality parameters. Here, Ω_{loc}^{+} and Ω_{conc}^{+} are the population counterparts of $\hat{\Omega}_{loc}$ and $\hat{\Omega}_{conc}$.

Confidence Regions

Theorem 5.3 (Asymptotic coverage $1 - \alpha$)

Under A1–A4, the confidence regions

$$\mathcal{C}_{global}^{1-\alpha} = \left\{ \mathbf{v} \in \mathbb{R}^q : n(\hat{\beta}_j - \mathbf{v})^\top \hat{\Sigma}_{\beta_j}^{-1} (\hat{\beta}_j - \mathbf{v}) \leq \chi_{q,\alpha}^2 \right\}$$

$$\mathcal{C}_{loc}^{1-\alpha} = \left\{ \mathbf{v} \in \mathbb{R}^q : n(\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j - \mathbf{v})^\top \hat{\Omega}_{loc}^+ (\mathbf{P}_{\hat{\mu}^\perp} \hat{\beta}_j - \mathbf{v}) \leq \chi_{q-1,\alpha}^2 \right\}$$

$$\mathcal{C}_{conc}^{1-\alpha} = \left\{ \mathbf{v} \in \mathbb{R}^q : n(\mathbf{P}_{\hat{\mu}} \hat{\beta}_j - \mathbf{v})^\top \hat{\Omega}_{conc}^+ (\mathbf{P}_{\hat{\mu}} \hat{\beta}_j - \mathbf{v}) \leq \chi_{1,\alpha}^2 \right\}$$

has asymptotic coverage $1 - \alpha$ for the targets $\beta_{j,0}, \mathbf{P}_{\mu_0^\top} \beta_{j,0}, \mathbf{P}_{\mu_0} \beta_{j,0}$.

- $\mathcal{C}_{global}^{1-\alpha}$ is a q -dimensional ellipsoid, since $\hat{\Sigma}_{\beta_j}$ has full rank. The other two precision matrices have rank $q - 1$ and 1, so each region constrains $\mathbf{v}$ only through its own projection.
- Therefore, $\mathcal{C}_{loc}^{1-\alpha}$ and $\mathcal{C}_{conc}^{1-\alpha}$ can be displayed as a $(q - 1)$ -dimensional ellipsoid on $\hat{\mu}^\perp$ and as an interval along $\hat{\mu}$, respectively.

Geometry of the location and concentration effects

- The two effects are the **direction** and the **length** of the natural-parameter curve with a single covariate x

$$\theta(x) = \mu + x\beta, \quad \underbrace{\zeta(x) = \frac{\theta(x)}{\|\theta(x)\|}}_{\text{direction}}, \quad \underbrace{\kappa(x) = \|\theta(x)\|}_{\text{length}}.$$

$\beta \parallel \mu$: pure concentration

$\theta(x) = (1 + xc)\mu$, so
 $\zeta(x) \equiv \frac{\mu}{\|\mu\|}$ and $\kappa(x) = |1 + xc| \|\mu\|$.
 Direction fixed, length varies; a covariate acts on conc. alone.

$\beta \perp \mu$: contaminated location

$\zeta(x) = (\mu + x\beta)/\kappa(x)$ and
 $\kappa(x)^2 = \|\mu\|^2 + x^2 \|\beta\|^2$.
 Direction rotates, but the length also grows by $x^2 \|\beta\|^2$ (**contaminated**).

- For a general β , one further **discrepancy** arises on the direction side, of angle $x^2 \|\mathbf{P}_\mu \beta\| \|\mathbf{P}_{\mu^\perp} \beta\| / \|\mu\|^2$, which vanishes whenever either component is zero.
- Both discrepancies decrease as $\|\mu\|$ **grows**, x **shrinks**, or $\|\beta\|$ **shrinks**. When this happens, $\mathbf{P}_{\mu^\perp} \beta$ and $\mathbf{P}_\mu \beta$ are the **pure** location and conc. effects.

How pure are the two effects in practice?

- Since both discrepancies are of second order in $x\beta$ relative to μ , each of them is bounded by the single quantity

$$\hat{r}_j = \frac{\max_i |x_{ij}| \cdot \|\hat{\beta}_j\|}{\|\hat{\mu}\|}.$$

$\hat{r}_j$	Max. Direction Gap	Max. Change in κ	Reading
0.1	0.3°	0.5%	Essentially pure
0.3	2.6°	4.4%	Borderline
0.5	7.2°	11.8%	Not pure

Max. Gap is bounded above by $\hat{r}_j^2/2$

Change in κ is measured relative to $\|\hat{\mu}\|$

- We note that the inference is valid for any $\hat{r}_j$; what changes is only how we describe the targets.

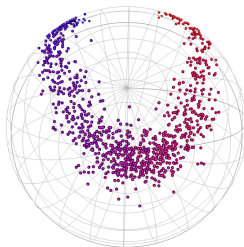
Nonlinear Extension: Quadratic Example

- Given a single covariate $x_i \in \mathbb{R}$, we can further consider orthogonal quadratic term $P_2(x_i)$ obtained by Gram–Schmidt on $\{1, x, x^2\}$:

$$\theta_i = \mu + x_i\beta_1 + P_2(x_i)\beta_2,$$

$$\text{subject to } \sum_i x_i = 0, \sum_i P_2(x_i) = 0, \sum_i x_i P_2(x_i) = 0$$

- Since this model is still linear in the parameters, it remains a canonical exponential family GLM, and the theory still holds as long as the assumptions are satisfied.



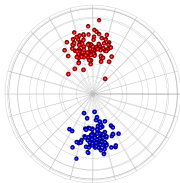
Simulation design

- $q = 3$, $\boldsymbol{\mu}_0 = (0, 0, 10)^\top$, $n \in \{50, 100, 200, 500\}$ with $n_1 = n_2 = n/2$.
- The binary X_1 and the continuous X_2 are generated from correlated normals with $\rho = 0.8$.
- With $\boldsymbol{\theta}_i = \boldsymbol{\mu}_0 + x_{i1}\boldsymbol{\beta}_1 + x_{i2}\boldsymbol{\beta}_2$ and $\mathbf{y}_i \mid \mathbf{x}_i \sim \text{vMF}(\boldsymbol{\theta}_i)$,

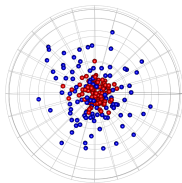
$$\boldsymbol{\beta}_j = s_{\text{loc},j} \mathbf{v}_{\perp,j} + s_{\text{conc},j} \frac{\boldsymbol{\mu}_0}{\|\boldsymbol{\mu}_0\|}, \quad \mathbf{v}_{\perp,j} \perp \boldsymbol{\mu}_0, \quad \|\mathbf{v}_{\perp,j}\| = 1.$$

with $j \in \{1, 2\}$. $s_{\text{loc},j}$ and $s_{\text{conc},j}$ control the location and conc. sizes.

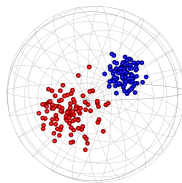
- We test X_1 with $s_{\text{loc},1} \in \{0, 1, 2\}$ and $s_{\text{conc},1} \in \{0, 2, 4\}$, and the confounder X_2 is set to either the null level $(0, 0)$ or the strong level $(2, 4)$.



(a) Location



(b) Concentration



(c) Combined

What we compare against

- (A) **Spherical regression** which can test global effect only.
 - ▶ **IAG, ESAG, SPherical Cauchy, PKBD** (in R package `Directional`).
 - ▶ Test the whole effect $H_0 : \beta_{1,0} = \mathbf{0}$ by LRT, not location and concentration separately.
- (B) **Effect-separating tests** for location and concentration effects.
 - ▶ **vMF-LRT** (Mardia, 2009): two-sample likelihood ratio test¹ for direction (χ^2_{q-1}) and concentration (χ^2_1), after two-stage studentization to adjust X_2 : (i) regress on X_2 for location, (ii) then regress $\log \|\text{residual}\|$ on X_2 for concentration.
 - ▶ **TS-REG**: Apply the regression based location and dispersion testing method after Tangent Space (TS) based euclideanization.
 - ▶ **TS-DGLM**: Apply the double GLM using separate link functions for location and dispersion, after Tangent Space (TS) based euclideanization.

¹ K. V. Mardia et al. John Wiley & Sons, 2009.

Type I Error Rate

- Each point is one effect size configuration of X_1 and X_2 under which the corresponding null holds.
 - Among the spherical regression methods, ESAG and PKBD fail to control the error rate,
 - Among the effect-separating methods, every competitor fails at some effect size configurations.

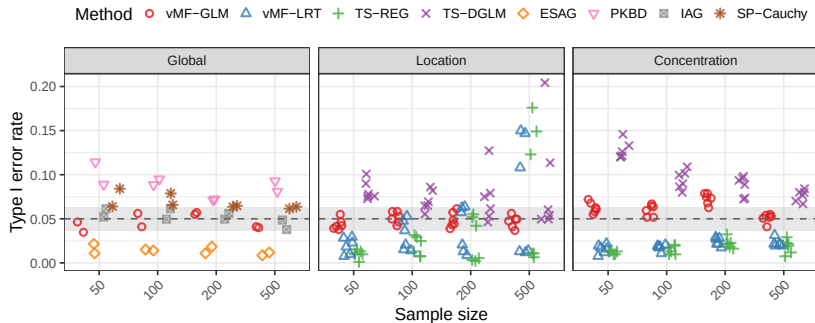


Figure 6: Empirical type I error rates for the binary covariate X_1 .

Statistical Power

Against Spherical Regressions (IAG, ESAG, PKBD, SP-Cauchy)

- The **horizontal** axis nests (i) the effect size of X_1 , (ii) the sample size, and (iii) the effect size of the confounder, and the **vertical** axis is the power.
- Except at a few configurations with small sample sizes, vMF-GLM attains the highest power.

Global

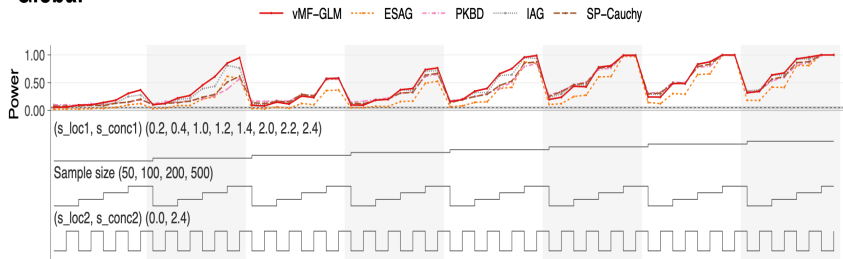
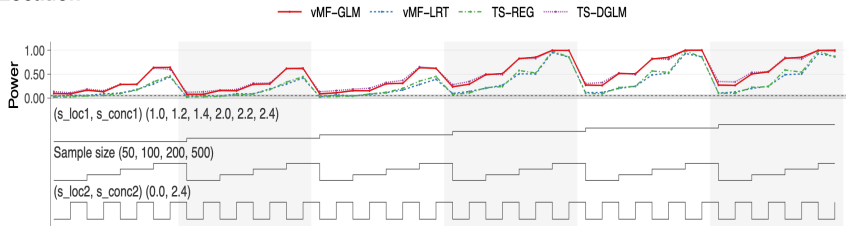


Figure 7: Empirical power for the binary covariate X_1 . The x-axis consists of three simulation levels (effect size of X_1 , the sample size, and the effect size of X_2).

Statistical Power

Against Effect-separating Methods (vMF-LRT, TS-REG, TS-DGLM)

Location



Concentration

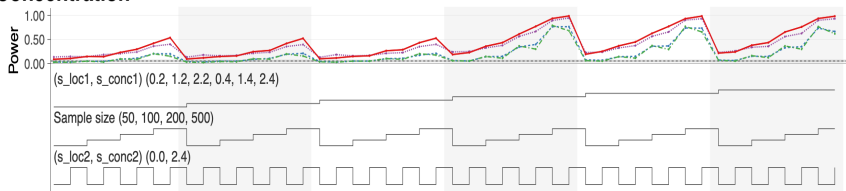


Figure 8: Empirical power for the binary covariate X_1 . The x-axis consists of three simulation levels (effect size of X_1 , the sample size, and the effect size of X_2).

Empirical Coverage Probabilities

with strong confounding

Table 1: Empirical coverage probabilities of the three confidence regions for the first covariate effect β_1 , with strong confounding effects.

		$n = 50$			$n = 100$			$n = 500$		
$s_{\text{loc},1}$	$s_{\text{conc},1}$	$\mathcal{C}_{\text{global}}$	$\mathcal{C}_{\text{loc}}$	$\mathcal{C}_{\text{conc}}$	$\mathcal{C}_{\text{global}}$	$\mathcal{C}_{\text{loc}}$	$\mathcal{C}_{\text{conc}}$	$\mathcal{C}_{\text{global}}$	$\mathcal{C}_{\text{loc}}$	$\mathcal{C}_{\text{conc}}$
0	0	0.955	0.950	0.955	0.954	0.954	0.955	0.953	0.953	0.946
0	2	0.956	0.951	0.945	0.952	0.955	0.947	0.947	0.951	0.949
0	4	0.958	0.954	0.951	0.952	0.955	0.944	0.944	0.946	0.950
1	0	0.961	0.961	0.948	0.955	0.953	0.947	0.952	0.958	0.950
1	2	0.958	0.956	0.951	0.950	0.947	0.947	0.947	0.951	0.945
1	4	0.957	0.955	0.948	0.951	0.950	0.950	0.940	0.946	0.944
2	0	0.960	0.955	0.952	0.954	0.953	0.950	0.947	0.949	0.946
2	2	0.956	0.954	0.949	0.951	0.953	0.945	0.948	0.943	0.950
2	4	0.957	0.956	0.955	0.951	0.947	0.950	0.942	0.941	0.945

Application to the ParkWest study

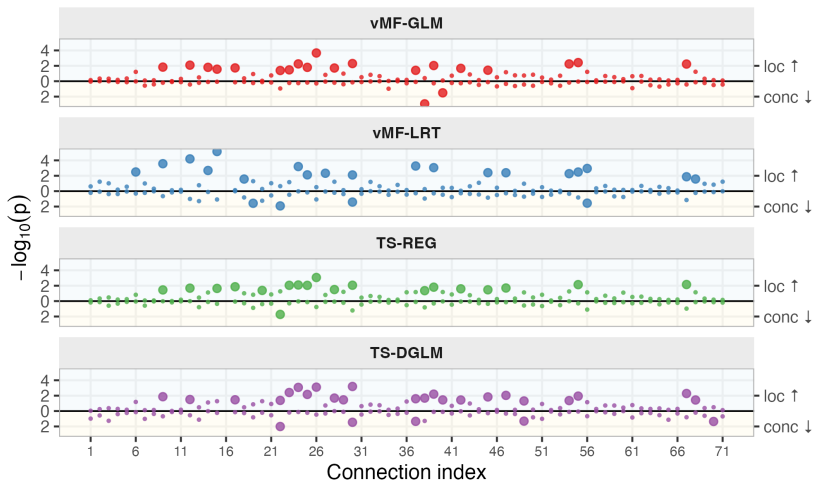
- The hippocampal surface of each subject is summarized by an LP-ds-rep with 70 connection directions $\mathbf{y}_{ik} \in \mathbb{S}^2$ for 182 patients and 108 controls.
- We fit the model separately at each connection $k = 1, \dots, 70$,

$$\theta_{ik} = \mu_k + \beta_{k,\text{grp}} \text{group}_i + \beta_{k,\text{len}} \text{len}_{ik} + \beta_{k,\text{age}} \text{age}_i + \beta_{k,\text{sex}} \text{sex}_i,$$

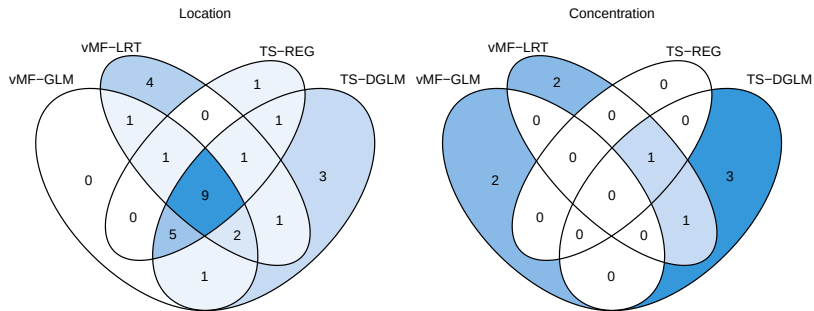
with all covariates centered at the connection so that $\sum_i \mathbf{x}_{ik} = \mathbf{0}$.

- The parameter of interest is $\beta_{k,\text{grp}}$, and the three remaining covariates are potential confounders.
- We report connection-level raw p -values without multiple correction.

Miami Plots for $-\log_{10} p\text{-value}$ under the adjusted Model



Venn Diagrams for Significant Frames under the adjusted Model



Discussion

Limitation and extension to the polysphere

- **Hemisphere restriction.** θ_i moves on $\mu + \text{span}(\beta_1, \dots, \beta_p)$, of dimension $\text{rank}(\mathbf{B})$.
 - ▶ $\text{rank}(\mathbf{B}) < q$: the mean directions stay in **one open hemisphere**
 - ▶ $\text{rank}(\mathbf{B}) = q$ (needs $p \geq q$): **the whole sphere** is reachable
- **Polysphere.** A response on $(\mathbb{S}^2 \times \mathbb{R}_+)^{70}$ compares the shapes themselves.
 - ▶ Beyond the spatial dependence of Lan et al. (2025), the LP-ds-rep carries a **tree structure** joining each frame to its parent.

Thank you for your attention ! 😊